



## RMV

3-way and 4-way rotary mixing valves  
DN20–DN50 · Kvs 2.5–40 m<sup>3</sup>/h · PN 10



[termojet.com.ua](http://termojet.com.ua)

**WARNING!** Read this manual before starting installation. For safety reasons, only persons familiar with its contents may install or service the product.

## Application

The **3-way** mixing and diverting valve is intended for heating, cooling and domestic hot water systems. It is most often used for mixing: the required flow temperature is obtained by blending return water into the hot water from the boiler. The same valve can also work as a diverting or a changeover valve.

Besides controlling the flow temperature, the **4-way** valve **raises the return temperature to the boiler** — this protects the boiler against low-temperature corrosion.

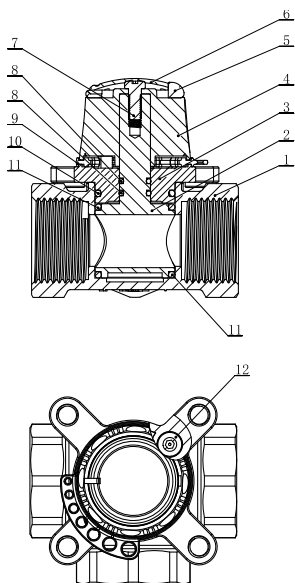
The valve can be operated by hand with its handle or electrically, with a Termojet AQUA series actuator; together they form a compact assembly.



3-way valve



4-way valve



## Parts

1. Valve body
2. Rotor
3. Cover
4. Handle
5. Handle position indicator
6. Concealed handle screw
7. Screw
8. O-ring
9. Position indicator
10. Cover screw
11. O-ring
12. Cover screw

Fig. 1. Valve construction

## Technical data

|                                |                                      |
|--------------------------------|--------------------------------------|
| Maximum working pressure       | PN 10 bar                            |
| Nominal bore                   | DN 20–50                             |
| Flow coefficient Kvs           | 2.5–40 m³/h                          |
| Working temperature range      | –10...110 °C                         |
| Operating torque               | DN20–DN32 < 1 N·m; DN40–DN50 < 3 N·m |
| Leakage in the closed position | < 0.1 %                              |
| Threaded connection            | EN 10226-1                           |
| Body, rotor, cover             | EN 12165 CW617N brass                |
| Handle, position indicator     | nylon                                |
| Seals                          | EPDM                                 |
| Medium                         | water, water with up to 50 % glycol  |

## Range

| Part no.     | Type  | DN | Kvs, m³/h | Thread |
|--------------|-------|----|-----------|--------|
| RMV03110-034 | 3-way | 20 | 2.5       | male   |
| RMV03100-034 | 3-way | 20 | 6.3       | female |
| RMV03110-100 | 3-way | 25 | 6         | male   |
| RMV03100-100 | 3-way | 25 | 10        | female |
| RMV03110-114 | 3-way | 32 | 12        | male   |
| RMV03100-114 | 3-way | 32 | 16        | female |
| RMV03110-112 | 3-way | 40 | 18        | male   |
| RMV03100-112 | 3-way | 40 | 25        | female |
| RMV03100-200 | 3-way | 50 | 40        | female |
| RMV04110-100 | 4-way | 25 | 12        | female |
| RMV04110-114 | 4-way | 32 | 18        | female |

# Dimensions

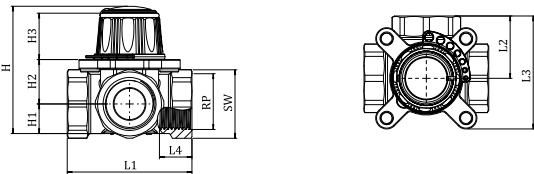


Fig. 2. 3-way valve

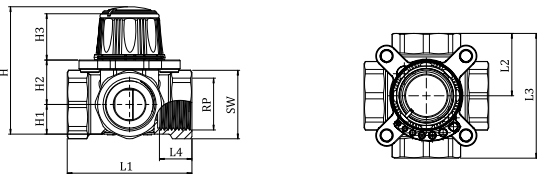


Fig. 3. 4-way valve

| DN | SW | RP     | L1  | L2   | L3 (3-w) | L3 (4-w) | L4   | H   | H1 | H2   | H3 |
|----|----|--------|-----|------|----------|----------|------|-----|----|------|----|
| 20 | 34 | 3/4"   | 80  | 40   | 72.5     | 80       | 16.5 | 82  | 19 | 28.5 | 28 |
| 25 | 42 | 1"     | 80  | 40   | 72.5     | 80       | 21   | 82  | 19 | 28.5 | 28 |
| 32 | 49 | 1 1/4" | 90  | 45   | 77.5     | 90       | 21   | 90  | 23 | 32.5 | 28 |
| 40 | 55 | 1 1/2" | 115 | 57.5 | 88.5     | —        | 26   | 101 | 28 | 39   | 28 |
| 50 | 66 | 2"     | 125 | 62.5 | 93.5     | —        | 22.5 | 101 | 28 | 39   | 28 |

Dimensions in millimetres. SW — across flats, RP — connection thread. 4-way valves are made in DN25 and DN32.

# Piping examples

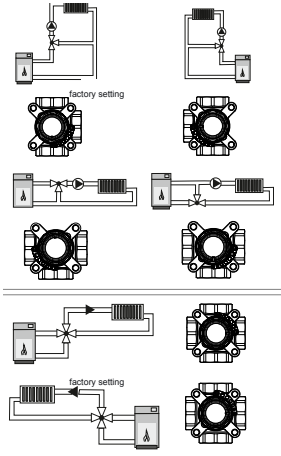


Fig. 4. Typical layouts with a 3-way and a 4-way valve: the matching rotor position is shown alongside

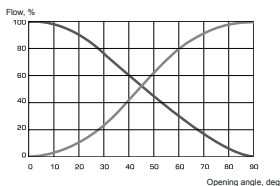


Fig. 5. Flow through each inlet against the rotor angle

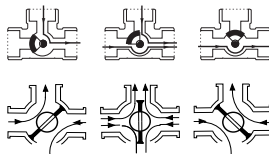


Fig. 6. Flow paths: 3-way valve above, 4-way valve below

## Changing the configuration

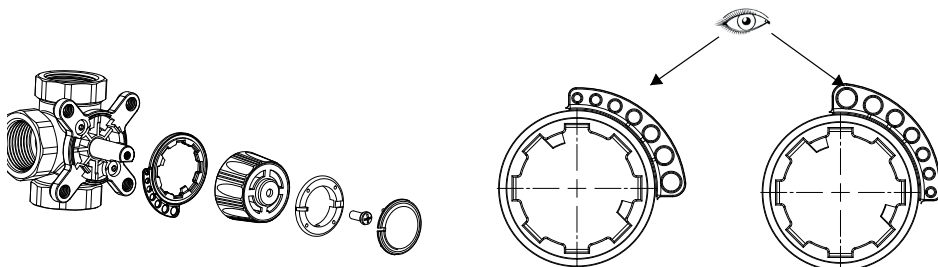


Fig. 7. Turning the indicator ring over and repositioning the handle changes the side on which the flow is connected

## Installation notes

- Installation is carried out by qualified trades.
- Before fitting, clean the pipework of scale and rust; a strainer is required at the system inlet.
- Fit the valve so that the direction of flow matches the chosen piping layout; the valve must remain accessible for servicing.
- To fit an actuator, remove the handle of the valve, secure the anti-rotation plate on the cover, fit the adapter onto the rotor spindle and turn its indicator to the 45° position — between marks 4 and 6. Then follow the manual of the AQUA series actuator.
- After installation, test the system for leaks.

Using pipe wrenches during installation is not permitted. Misalignment of the ports and uneven tightening of the threads are not permitted either.

## Product data

|                        |                                                                                     |
|------------------------|-------------------------------------------------------------------------------------|
| Product                | Termojet RMV rotary mixing valve                                                    |
| Manufacturer           | Termojet                                                                            |
| Distributor in Ukraine | Sofiivka-Montazh LLC, EDRPOU 37074476<br>EDRPOU — Ukrainian company registry code   |
| Address                | 3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine |
| Contacts               | +380 44 407 24 10 ·<br>termojet@sofiievka.kiev.ua<br>termojet.com.ua                |

## Storage and transport

Store the product in the manufacturer's packaging, in heated or unheated warehouses — no closer than one metre to any heating appliance — or under a canopy, in conditions that rule out mechanical damage.

Permissible storage and transport temperature is –20 to +50 °C. The product may be carried by any means of transport in accordance with the carriage rules in force for that mode of transport. During carriage, protect the product from impact, mechanical load and surface scratching.

## Disposal

The product contains no substances hazardous to human life and health or to the environment. At the end of its service life, dispose of the product as household waste in accordance with local waste-management rules. Metal parts are suitable for remelting.

## Acceptance and testing

The products covered by these instructions have been manufactured, tested and accepted in accordance with the manufacturer's current technical documentation.

## Warranty

The manufacturer warrants that the product complies with safety requirements, provided that the user observes the rules set out in these instructions.

The service life of the product, provided that these instructions are observed and the required servicing is carried out, is **10 years** from the date the product is handed over to the user.

The warranty period is **24 months** from the date of sale, but no longer than the service life of the product.

The warranty covers all defects arising through the fault of the manufacturer and **does not cover** defects caused by:

- failure to observe the storage, installation, testing, operation or servicing conditions described here;
- improper transport or loading and unloading;
- exposure to substances aggressive to the materials of the product;
- fire, natural disaster or force majeure;
- acts of the user or third-party interference with the design of the product.

Products that fail within the warranty period because of a manufacturing defect are repaired or replaced free of charge. The cost of removing and transporting the faulty product is not reimbursed to the buyer. If a claim proves to be unfounded, the buyer pays for the diagnosis and examination.

To process a claim the following are required: a free-form statement giving contact details, the installation address and a description of the defect; proof of purchase; photographs of the faulty product, including photographs taken at the place of installation; a copy of the completed warranty card.

Claims are accepted at: 3 Kyivska St., Sofiivska Borshchahivka, Bucha district, Kyiv region, 08131, Ukraine. Phone +380 44 407 24 10, e-mail [termojet@sofiievka.kiev.ua](mailto:termojet@sofiievka.kiev.ua).

The manufacturer reserves the right to introduce changes to the design of the «Termojet RMV rotary mixing valve» product that do not impair its quality.

Warranty card

to delivery note No. \_\_\_\_\_ dated \_\_\_\_\_ 20\_\_\_\_

Product: Termojet RMV rotary mixing valve

| No. | Part number | Quantity | Notes |
|-----|-------------|----------|-------|
| 1   |             |          |       |
| 2   |             |          |       |
| 3   |             |          |       |
| 4   |             |          |       |

Warranty period – 24 months from the date of sale.

I have read and accept the warranty terms and the installation and operating instructions:

\_\_\_\_\_  
Buyer (signature)

\_\_\_\_\_  
Seller (signature)

\_\_\_\_\_  
Date of sale

Stamp or seal  
of the selling organisation